

NAGPUR AMBEDKAR INTL, India

WMO: 428670

Lat: 21.092N

Lon: 79.047E

Elev: 1033

StdP: 14.16

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.1	55.1	32.7	28.3	84.5	35.5	31.6	81.0	12.2	77.7	10.8	77.1	1.7	340	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.6	111.6	72.6	109.6	72.4	107.6	72.1	82.1	90.2	81.0	89.4	80.1	88.4	7.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	164.9	86.1	79.1	156.7	84.6	78.5	153.6	84.0	47.2	90.9	45.9	88.8	44.9	89.2	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.5	11.7	DB	46.4	115.2	2.6	2.3	44.5	116.9	43.0	118.3	41.6	119.6	39.7	121.3	(4)
(5)			WB	42.6	84.9	2.6	3.1	40.7	87.2	39.2	89.0	37.8	90.7	35.9	93.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	70.1	75.8	83.7	91.8	96.5	90.0	82.3	81.1	82.0	80.1	75.0	70.3	(6)	
(7)		DBStd	8.82	3.82	4.24	4.69	4.06	4.12	6.58	3.35	2.53	2.28	3.03	3.32	3.59	(7)	
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD65	9	5	0	0	0	0	0	0	0	0	0	0	4	(9)	
(10)		CDD50	11524	623	721	1045	1255	1442	1200	1001	964	961	934	749	629	(10)	
(11)		CDD65	6060	163	302	580	805	977	750	536	499	511	469	299	169	(11)	
(12)		CDH74	77081	1572	3206	7781	12912	16633	10883	5603	4703	5026	4545	2647	1570	(12)	
(13)	CDH80	43737	472	1534	4703	8888	12227	6882	2058	1441	1947	2090	1048	447	(13)		
(14)	Wind	WSAvg	4.5	2.9	3.6	4.0	5.1	6.7	7.1	6.5	5.7	4.3	2.8	2.3	2.4	(14)	
(15)	Precipitation	PrecAvg	44.4	0.5	0.6	0.7	0.4	0.8	7.1	12.6	11.0	7.2	2.0	0.6	0.5	(15)	
(16)		PrecMax	69.1	4.1	3.5	3.9	2.4	3.9	18.0	26.7	22.0	16.8	14.3	3.9	6.5	(16)	
(17)		PrecMin	23.9	0.0	0.0	0.0	0.0	0.0	0.9	4.1	3.1	0.3	0.0	0.0	0.0	(17)	
(18)		PrecStd	9.1	0.7	0.7	1.0	0.5	0.9	3.8	4.8	4.1	3.9	2.7	0.9	1.4	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	98.5	106.2	112.7	116.7	113.4	97.2	93.0	95.2	96.5	91.8	89.6	(19)	
(20)			MCWB	67.4	66.4	69.2	70.4	73.2	73.3	78.6	81.1	78.0	74.8	69.7	67.9	(20)	
(21)		2%	DB	87.4	94.8	103.8	109.8	113.1	109.5	94.0	90.7	93.0	93.7	89.7	86.4	(21)	
(22)			MCWB	65.4	66.9	68.0	70.2	73.2	74.7	78.5	79.8	78.7	74.6	70.1	66.7	(22)	
(23)		5%	DB	84.6	91.7	100.8	108.0	111.1	106.0	91.3	88.7	91.3	91.8	88.0	84.5	(23)	
(24)			MCWB	64.3	66.5	67.8	70.2	73.1	75.5	78.8	78.9	78.3	74.2	69.8	65.0	(24)	
(25)		10%	DB	82.5	89.3	98.4	105.9	109.1	102.2	88.9	87.1	89.5	89.9	86.2	82.4	(25)	
(26)			MCWB	63.6	65.6	67.6	70.1	73.0	75.5	78.6	78.5	78.0	73.8	68.9	63.9	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	73.4	75.6	80.2	82.0	83.2	83.6	83.6	82.6	81.0	77.5	73.5	(27)
(28)				MCD B	84.4	86.0	93.4	99.5	105.4	93.6	89.4	89.6	90.1	89.1	86.3	83.9	(28)
(29)	2%		WB	69.1	70.9	73.6	77.8	79.5	81.3	81.6	81.5	81.0	79.3	75.5	70.5	(29)	
(30)			MCD B	80.6	85.2	92.8	99.0	103.1	92.0	87.2	87.3	88.4	87.4	84.0	81.0	(30)	
(31)	5%		WB	67.3	69.1	71.9	75.3	77.7	80.0	80.4	80.2	80.0	78.2	73.6	68.1	(31)	
(32)			MCD B	79.9	84.5	91.6	97.0	101.3	91.3	86.5	85.9	87.2	86.0	82.2	78.5	(32)	
(33)	10%		WB	65.5	67.6	70.3	73.3	76.3	79.0	79.4	79.2	79.1	76.9	71.8	66.4	(33)	
(34)			MCD B	78.4	84.4	90.4	97.0	99.4	89.9	85.3	84.5	86.1	84.5	80.0	77.6	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	25.9	26.5	27.6	27.8	24.6	18.1	11.3	10.7	14.3	20.8	24.4	26.2	(35)
(36)				MCDBR	28.2	29.6	31.1	29.4	26.1	22.9	15.7	14.1	18.0	23.8	25.7	28.2	(36)
(37)		MCWBR		12.0	11.4	11.0	9.9	9.0	7.0	5.7	6.3	7.3	9.2	10.4	12.1	(37)	
(38)		5% WB	MCD BR	24.6	25.3	26.2	26.9	24.9	18.1	12.9	12.6	15.2	17.3	19.8	23.1	(38)	
(39)			MCWBR	12.8	11.8	10.6	11.1	9.8	7.0	6.1	6.5	7.2	8.0	10.1	12.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.463	0.478	0.520	0.552	0.584	0.635	0.591	0.555	0.540	0.517	0.507	0.463	(40)		
(41)		taud	2.029	2.012	1.935	1.902	1.861	1.797	1.906	2.014	2.035	2.032	1.980	2.048	(41)		
(42)		Ebn at Noon	252	258	252	246	236	222	233	242	244	244	237	247	(42)		
(43)		Edh at Noon	50	54	61	64	66	69	62	56	54	52	53	48	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1449	1718	1987	2193	2195	1650	1174	1203	1522	1645	1487	1401	(44)		
(45)		RadStd	69	76	100	69	55	171	165	133	137	99	79	53	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.85	N/A	N/A	+1.39	+1.55	+1.37	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page